

Week 1 module

C and C++ Programming

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C programming

① C Syntax and Basic Structure

C is a procedural and structured programming language. Every standard C program has a precise, structured layout.

Header files (`#include <stdio.h>`)

This pre-processor directive includes the standard Input/output library. It allows the program to use the core functions, such as `printf()` for output and `scanf()` for input.

Main function (`int main()`)

This is the ~~most~~ mandatory entry point of any C program. Execution starts at the opening brace `{` of `main()`. Execution ends at the terminating semicolon of `return()`.

Output function (`printf`)

Unlike C++, C uses `printf()` function to display text on the screen. Double quotes let you pass format specifiers or escape sequences like `\n` (newline).

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Semicolon (;) : Every statement in C must end with a semicolon. It is a statement terminator for the compiler.

② Conditions and loops

Conditional statements (if-else) : Used to make decisions. The if block runs when the condition is true. If it is false, the program skips it or executes the optional else block.

Loops (for loop) : To repeat a block of code a certain number of times. It is composed of initialization, condition checking and increment/decrement expressions.

Nested loops ~~for~~ for star patterns to print two dimensional grid patterns we need a loop in another loop.

Outer loop : Manages the rows (vertical progression)

Inner loop : Handles the columns (prints spaces or stars horizontally for the current flow).

③ Programming problems

Solve the given star pattern Questions

① square

O/p: →

```
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *
```

```
#include <stdio.h>
```

```
int main () {
```

```
    int n=5; // total rows and columns
```

```
    // outer loop for rows
```

```
    for (int i=1; i<=n; i++) {
```

```
        // Inner loop for columns
```

```
        for (int j=1; j<=n; j++) {
```

```
            printf("*");
```

```
        }  
        printf("\n"); // move to the next line
```

```
    }  
    return 0;
```


② Hollow square pattern

```
#include <stdio.h>
int main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            // print stars only at the boundaries
            if (i == 1 || i == n || j == 1 || j == n) {
                printf("*");
            } else {
                printf(" ");
            }
        }
        printf("\n");
    }
    return 0;
}
```

O/p!:-

```
* * * * *
*       *
*       *
*       *
* * * * *
```

3) Triangle

```
#include <stdio.h>
int main()
{
    int n=5;
    for (int i=1; i<=n; i++)
    {
        // inner loop runs up to the current row number i
        for (int j=1; j<=i; j++)
        {
            printf("%d", j);
        }
        printf("\n");
    }
    return 0;
}
```

O/p:-

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

④ Inverted Triangle

```
#include <stdio.h>
```

```
int main ()
```

```
int n = 5;
```

Outer loop counts down from 'n' to 1

```
for (int i = n; i >= 1; i--) {
```

for (int j = 1; j <= i; j++) {

```
print f("d");
```

3

```
printf("\n");
```

2

```
return 0;
```

3

O/P:-

木 木 木 木 木

天 天 天 天

4 2 6

水 子

5) Pyramid

```
#include <stdio.h>
int main () {
    int n=5;
    for (int i = 1; i <= n; i++) {
        // print leading spaces
        for (int j = 1; j <= n - i; j++) {
            printf(" ");
        }
        // print stars based on odd number formula
        for (int k = 1; k <= (2 * i - 1); k++) {
            printf(" *");
        }
        printf("\n");
    }
    return 0;
}
```

O/p! — *

```

  * * *
 * * * * *
* * * * * *
* * * * * * *

```


⑥ Inverted pyramid

```
#include <stdio.h>
int main()
{
    int n = 5;
    // outer loop handles rows in reverse order
    for (int i = n; i >= 1; i--) {
        // print spaces
        for (int j = 1; j <= n - i; j++) {
            printf(" ");
        }
        // print stars
        for (int k = 1; k <= (2 * i - 1); k++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

O/p:-

```

* * * * *
 * * * * 
  * * *  
   * *   
    *
  
```


Q7) Half diamond

```
#include <stdio.h>
int main() {
    int n = 5;
    // Upper half: Growing Triangle
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf(" * ");
        }
        printf("\n");
    }
    // Lower half: Shrinking Inverted Triangle
    for (int i = n - 1; i >= 1; i--) {
        for (int j = 1; j <= i; j++) {
            printf(" * ");
        }
        printf("\n");
    }
    return 0;
}
```

o/p:-

```

 *
 * *
 * * *
 * * * *
 * * * * *
 * * * *
 * * *
 * *
 *
```

② Diamond

```
#include <stdio.h>
int main () {
    int n=5;
    // Upper Half: Regular pyramid
    for (int i=1; i<=n; i++) {
        for (int j=1; j<=n-i+1; j++) {
            printf(" ");
        }
        for (int k=1; k<=(2*i-1); k++) {
            printf("%d");
        }
        printf("\n");
    }
    // Lower Half: Inverted pyramid
    for (int i=n-1; i>=1; i--) {
        for (int j=1; j<=n-i+1; j++) {
            printf(" ");
        }
        for (int k=1; k<=(2*i-1); k++) {
            printf("%d");
        }
        printf("\n");
    }
    return 0;
}
```

O/p:-

```

      1
     22
    333
   4444
  55555
 44444
 33333
 22222
 11111

```


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Language: C

main.c

```
1 #include <stdio.h>
2
3 int main() {
4     int n = 5; // Total rows and columns
5     // Outer loop for rows
6     for (int i = 1; i <= n; i++) {
7         // Inner loop for columns
8         for (int j = 1; j <= n; j++) {
9             printf("=");
10        }
11        printf("\n"); // Move to the next line
12    }
13    return 0;
14 }
```

input

```
*****
*****
*****
*****
*****

...Program finished with exit code 0
Press ENTER to exit console.
```

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Toolbar: Run, Debug, Stop, Share, Save, Beauty, Language: C

main.c

```
1 #include <stdio.h>
2
3 int main() {
4     int n = 5;
5     for (int i = 1; i <= n; i++) {
6         for (int j = 1; j <= n; j++) {
7             // Print star only at the boundaries
8             if (i == 1 || i == n || j == 1 || j == n) {
9                 printf("*");
10            } else {
11                printf(" "); // Print empty space inside
12            }
13        }
14        printf("\n");
15    }
16    return 0;
17 }
```

input

```
*****
* *
* *
* *
*****

...Program finished with exit code 0
Press ENTER to exit console.
```

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main.c

```
1 #include <stdio.h>
2
3 int main() {
4     int n = 5;
5     for (int i = 1; i <= n; i++) {
6         // Inner loop runs up to the current row number 'i'
7         for (int j = 1; j <= i; j++) {
8             printf("*");
9         }
10        printf("\n");
11    }
12    return 0;
13 }
```

input

```
**
***
****
*****

...Program finished with exit code 0
Press ENTER to exit console.
```

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Address bar: https://www.onlinegdb.com/online_c_compiler#

Toolbar: Run, Debug, Stop, Share, Save, Beautify

Language: C

main.c

```
1 #include <stdio.h>
2
3 int main() {
4     int n = 5;
5     // Outer loop counts down from 'n' to 1
6     for (int i = n; i >= 1; i--) {
7         for (int j = 1; j <= i; j++) {
8             printf("*");
9         }
10        printf("\n");
11    }
12    return 0;
13 }
```

input

```
*****
****
***
**
*

...Program finished with exit code 0
Press ENTER to exit console.
```

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Language: C

main.c

```
1 #include <stdio.h>
2
3 int main() {
4     int n = 5;
5     for (int i = 1; i <= n; i++) {
6         // Print leading spaces
7         for (int j = 1; j <= n - i; j++) {
8             printf(" ");
9         }
10        // Print stars based on odd number formula: 2 * i - 1
11        for (int k = 1; k <= (2 * i - 1); k++) {
12            printf("*");
13        }
14        printf("\n");
15    }
16    return 0;
17 }
```

input

```
***
*****
*****
*****

...Program finished with exit code 0
Press ENTER to exit console.
```

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Language: C

main.c

```
1 #include <stdio.h>
2
3 int main() {
4     int n = 5;
5     // Outer loop handles rows in reverse order
6     for (int i = n; i >= 1; i--) {
7         // Print spaces
8         for (int j = 1; j <= n - i; j++) {
9             printf(" ");
10        }
11        // Print stars
12        for (int k = 1; k <= (2 * i - 1); k++) {
13            printf("*");
14        }
15        printf("\n");
16    }
17    return 0;
18 }
```

input

```
*****
*****
****
***
**
*

...Program finished with exit code 0
Press ENTER to exit console.
```

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Language: C

main.c

```
1 #include <stdio.h>
2
3 int main() {
4     int n = 5;
5     // Upper Half: Growing Triangle
6     for (int i = 1; i <= n; i++) {
7         for (int j = 1; j <= i; j++) {
8             printf("*");
9         }
10        printf("\n");
11    }
12    // Lower Half: Shrinking Inverted Triangle
13    for (int i = n - 1; i >= 1; i--) {
14        for (int j = 1; j <= i; j++) {
15            printf("*");
16        }
17        printf("\n");
18    }
19    return 0;
20 }
```

input

```
***
****
*****
*****
****
***
**
*

...Program finished with exit code 0
Press ENTER to exit console.
```

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Language: C

main.c

```
1 #include <stdio.h>
2
3 int main() {
4     int n = 5;
5     // Upper Half: Regular Pyramid
6     for (int i = 1; i <= n; i++) {
7         for (int j = 1; j <= n - i; j++) printf(" ");
8         for (int k = 1; k <= (2 * i - 1); k++) printf("*");
9         printf("\n");
10    }
11    // Lower Half: Inverted Pyramid
12    for (int i = n - 1; i >= 1; i--) {
13        for (int j = 1; j <= n - i; j++) printf(" ");
14        for (int k = 1; k <= (2 * i - 1); k++) printf("*");
15        printf("\n");
16    }
17    return 0;
18 }
```

input

```
***
*****
*****
*****
*****
***
*
```

...Program finished with exit code 0
Press ENTER to exit console.

https://www.onlinegdb.com/online_c_compiler#tab-stdin

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